C Programming Top 40 Viva Voce Questions & Answers

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1. Memory, Pointers & Array Viva Questions

- Q1: What is the difference between malloc() and calloc()?

  A: malloc() allocates uninitialized memory; calloc() allocates contiguous memory and zeroes all bits.

- Q2: Explain the difference between array name 'a' and '&a' in C.

  A: 'a' decays to a pointer to first element (int*); '&a' is a pointer to the entire array (int(*)[N]).

2. Functions & Structures Viva Questions

- Q3: What is the difference between Structure and Union in C?

  A: Structure allocates memory equal to sum of all members; Union shares memory equal to largest member.

- Q4: What is a Dangling Pointer and how can it be prevented?

  A: A pointer pointing to deallocated memory; prevent by assigning ptr = NULL immediately after free(ptr).